

Morgan Mathison

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Education

Bachelor of Science in In Plant Science

Summa Cum Laude

Emphasis in Breeding, Biology, and Biotechnology

May 2025

University of Missouri - Columbia

Minor in Biology, Certification in Retail Floral Management

GPA: 3.96

Research Experience

Postbaccalaureate NSF-RaMP Researcher

University of Kansas - Department of Evolution and Ecology

July 2025- June 2026

Laboratory of Dr. Brian Atkinson

Project: *Fossil Fruit Provides Direct Evidence for Cornales in Late Cretaceous Antarctica*

- Implemented cellulose acetate peeling methodology in order to prepare and section research fossil specimens for analysis
- Utilized brightfield microscopy to score and describe fossil Cornales fruit for integration into a morphological matrix
- Employed Mr. Bayes for conducting total-evidence Bayesian phylogenetic analyses to determine the phylogenetic affinity of fossil Cornales fruit

Undergraduate Research Assistant

University of Missouri - Division of Plant Science and Technology

May 2023-May 2025

Laboratory of Dr. Jacob Washburn

Thesis: *Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize*

- Piloted an unmanned aircraft system to collect hyperspectral and multispectral images of maize fields for further analysis and indice extraction
- Utilized Pix4D and Quantum Geographic Information System to stitch together over 10,000 individual images per drone flight for orthomosaic creation
- Operated Rstudio and Python to extract vegetative indices from orthomosaics to better understand maize health throughout the growing season

Publications in Progress

(Preprint) Mitchell, M., Sidberry, G., **Mathison, M.**, DeSalvio, A. J., Kick, D., & Washburn, J.

D. (2024). Drone-Based Identification of Flood-Tolerant Maize via Multispectral

Imaging: A Real-World Case Study. *bioRxiv*, 2024.11.26.625493.

<https://doi.org/10.1101/2024.11.26.625493>

Honors and Awards

University of Kansas Chancellors Doctoral Fellowship	Fall 2026-Present
National Science Foundation Graduate Research Fellowship Program - Honorable Mention	2026
University of Missouri - Dean's List	Spring and Fall 2022, 2023, 2024, Spring 2025
Missouri Seedmen's Association Honorary Member Scholarship (\$1000)	September 2024
University of Missouri Digital Agriculture Symposium - 3rd Place Best Poster	June 2024
Southeast Missouri State University - Dean's List	Fall 2021
Southeast Missouri State University - President's List	Fall 2021

Posters and Presentations

Oral Presentations

Mathison, M. (June, 2026). *Fossil Fruits Provide Direct Evidence for Cornales in the Late Cretaceous of Antarctica*. Konza Symposium 2026. Konza Prairie Biological Station, KS.

Mathison, M. (April, 2025). *Utilizing Drone Imaging to Monitor Crop Health in Maize*. University of Missouri Plant Science Senior Capstone Presentations. Columbia, MO.

Poster Presentations

Mathison, M. (April, 2026). *Fossil Fruit From the Late Cretaceous of Antarctica Sheds Light on the Early Biogeographic History of Cornales, The Dogwood Order*. Office For Advancing Success in Science Student Research Symposium, Lawrence, KS.

Mathison, M. (August, 2025.) *Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize*. Kansas Innovation Fest, Overland Park, KS.

Mathison, M. (April, 2025.) *Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize*. University of Missouri Show Me Research Week. Columbia, MO.

Mathison, M. (March, 2025.) *Utilizing Drone Imaging in Analysis of Genotype by Environment in Maize*. Maize Genetics Meeting 2025. St Louis, MO.

Mathison, M. (December, 2024.) *Utilizing Drone Imaging to Understand Plant Health*. University of Missouri Fall Undergraduate Research Day. Columbia, MO.

Mitchell, M., **Mathison, M.** (June, 2024.) *Drone Based Identification of Flood-Tolerant Maize Genotypes*. University of Missouri Digital Agriculture Symposium. Columbia, MO.

Mitchell, M., **Mathison, M.** (April, 2024.) *Drone Based Identification of Flood-Tolerant Maize Genotypes*. University of Missouri Show Me Research Week. Columbia, MO.

Mitchell, M., **Mathison, M.** (March, 2024.) *Drone Based Identification of Flood-Tolerant Maize Genotypes*. University of Missouri Plant Research Symposium. Columbia, MO.

Leadership Experience

Emporia State University Enhancing Your Future with Mathematics and Science “DIY Dino” Workshop

Co-Creator and Presenter

March 2026

- Designed suitable curriculum for a workshop for middle school students interested in STEM, showcasing the process of fossilization and how scientists use the fossil record to conduct research
- Facilitated activities with students handling fossils, providing a productive hands-on learning experience to individuals interested in Paleontology

Postbaccalaureate NSF-RaMP Researcher

Peer Education, Advancement, and Research in the Life Sciences (PEARLS)

Co-Creator

July 2025- June 2026

- Co-founded PEARLS, a peer cooperative supporting professional growth of postbaccalaureate researchers in the life sciences
- Co-organized “Graduate Experience Panel”, facilitating discussion between current graduate students and aspiring scholars in the life sciences

University of Missouri - Columbia

Transfer Experience and Advising Mentors

Director of Leadership Development

January - December 2024

- Wrote and sent weekly newsletters to over 400+ club members to inform students of opportunities happening on campus and ways to get involved
- Interacted with stakeholders during weekly meetings to understand club performance and identify ways to better support transfer students

Director of Communications

January - December 2023

- Created content for club and the University of Missouri Transfer Student Center related meetings and activities utilizing Canva for promotion of weekly meetings to increase member participation
- Planned and executed weekly club meetings alongside fellow executive members, providing a weekly space for transfer students to connect with one another

University of Missouri – Columbia

Center for Student Involvement

Welcome Week Leader

August 2023, August 2024

- Supervised over 125+ incoming transfer students during activities the week leading up to the start of fall classes to familiarize students to campus
- Coordinated activity involvement and facilitated participation to introduce new students to one another and build a strong system of connections within the transfer student community

References

Dr. Kelly K.S. Matsunaga

Graduate Research Advisor
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Dr. Brian Atkinson

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Dr. Jacob Washburn

Undergraduate Research Advisor
USDA ARS Plant Genetics Research Unit
University of Missouri
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Jacob.Washburn@usda.gov